# OSIRIS QED Extension: Thermal-Bath Axion Emission, Absorption, and Plasma Feedback

*Technical documentation for the axion module additions*

## 1. Scope and overview

This document describes the axion-physics extensions integrated into the OSIRIS QED framework. Only axion-related physics and numerics are documented; pre-existing QED capabilities (photon emission, pair creation, etc.) are assumed known and are not repeated here.

The extensions implement: (i) a massless axion particle species integrated into the existing QED particle pipelines; (ii) thermal-bath axion emission channels (electron-ion/electron-electron bremsstrahlung and Compton-like production); (iii) Primakoff conversion between photons and axions in a screened Coulomb field; (iv) inverse processes for absorption using enforced detailed balance; (v) conservative energy and momentum exchange between axions and the plasma through cell-local deposition buffers and a conservative feedback operator; (vi) optional local temperature evolution with self-consistent recomputation of emission and absorption rates; and (vii) variance-reduction and importance-sampling measures including controlled macroparticle weights and Poisson macro-event sampling with saturation avoidance.

## 2. Integration of an axion species into OSIRIS QED pipelines

Axions are represented as a massless particle species advanced using the same kinematic propagation machinery used for photons in the QED module. The implementation introduces a dedicated species type and a set of axion-process operators that are invoked within the established QED time-stepping sequence.

### 2.1 Axion species data structure

The axion species type extends the QED particle-species infrastructure (massless kinematics, boundary handling, merging/deletion, and diagnostics hooks) and is thus compatible with existing OSIRIS infrastructure for particle communication and restart.

```
Relevant definitions: os-axions-define.f03 :: module
m_axions_define, type t_axions (extends t_species_qed)
```

### 2.2 Axion propagation

Axions are treated as ultrarelativistic (massless) particles. In normalized OSIRIS units the axion momentum vector p satisfies |p| = E/(m_e c^2). The position update uses the characteristic velocity v = p/|p|.

*x^{n+1} = x^{n} + (p/|p|) Δt / Δx, with integer-cell trimming applied per direction.*

```
Propagation operators: os-axions.f03 :: advance_axions_1d /
advance_axions_2d / advance_axions_3d / advance_axions_cyl_2d
```

### 2.3 Placement in the time-stepping sequence

Axion creation occurs in two distinct contexts: (i) during the QED push of emitting charged species (thermal-bath bremsstrahlung and Compton-like production), where axion macroparticles are created directly into the axion species container; and (ii) during Primakoff conversion, where photons are converted into axions (and optionally the inverse). Axion absorption (inverse processes) is applied as an operator acting on the axion species, with rates obtained by detailed balance relative to the forward emissivity model.

```
Emission insertion points: os-spec-push-qed.f03 :: axion
bremsstrahlung (ax_model=2) and axion_compton_emission

Absorption operators: os-axions.f03 :: axion_bremsstrahlung_absorb,
axion_compton_absorb

Primakoff conversion: os-axion-primakoff.f03 ::
primakoff_photon_to_axion, primakoff_axion_to_photon
```

## 3. Thermal-bath axion emission processes

The thermal-bath emission channels implemented here are not binary-collision Monte Carlo operators between explicit particles. Instead, emission is sampled from a local emissivity model derived for a thermal plasma, evaluated using local density proxies and (optionally) an evolving local temperature field. Emission is represented by Poisson-distributed macro-events per timestep.

### 3.1 Common sampling architecture

For an emitting macroparticle with weight w_e (in units of real particles), the expected number of emitted real axions in a timestep is w_e Γ Δt, where Γ is the emission rate per real emitter. The algorithm samples a Poisson-distributed integer k with mean λ_macro equal to the expected number of emitted axion macroparticles in the timestep. Variance control is achieved by selecting a macroparticle weight w_a and defining λ_macro = (w_e Γ Δt)/w_a.

$$k \sim \mathrm{Poisson}(\lambda_{macro}), \quad \lambda_{macro} = (w_e \Gamma \Delta t) / w_a.$$

To avoid pathological saturation when λ_macro becomes large, a safety cap λ_cap is enforced. If λ_macro > λ_cap, the mean is reduced to λ_cap and the emitted macroparticle weight is scaled upward by w_scale = λ_macro/λ_cap so that the expectation value of emitted real axions is preserved.

$$\text{If } \lambda_{macro} > \lambda_{cap}: \ \lambda_{eff} = \lambda_{cap}, \ w_a \leftarrow w_a (\lambda_{macro}/\lambda_{cap}).$$

In all channels, axion energies are sampled from a proxy spectral distribution in the dimensionless variable w = E/T (with E in keV and T in keV), and directions are sampled isotropically unless otherwise stated by the process model.

## 3.2 Bremsstrahlung axion emission (thermal plasma model, ax_model=2)

The bremsstrahlung channel implements thermal-plasma axion production following a non-degenerate emissivity model. The implemented rate is evaluated locally using a proxy for the mass density obtained from an ion charge-density deposit and a user-specified ion mass number A_ion. Screening corrections and the optional electron-electron contribution are included as in the adopted emissivity parametrisation.

Let Q_a be the volumetric axion emissivity (erg cm^-3 s^-1). A number emissivity is obtained via division by a mean energy <E> = <w> k_B T, where <w> is the mean of the proxy w-distribution. The corresponding number-production rate per electron is Γ = (1/n_e) (ṅ_a), with n_e inferred from neutrality by default.

$\dot{n}_a = Q_a / \langle E \rangle, \quad \Gamma = \dot{n}_a / n_e.$

The proxy spectrum is represented as a Gamma distribution in w with fixed shape and scale parameters. Denote the probability density f(w). The emission sampler draws k macro-events and, for each event, samples w and sets the emitted axion energy E = w T. The dimensionless OSIRIS energy is E/(m_e c^2) = (w T_keV)/(511 keV). Directions are sampled isotropically.

$f(w) = [1/(\Gamma(k)\,\theta^k)]\, w^{k-1} \exp(-w/\theta), \quad E_{dimless} = (w\, T_{keV})/511.$

```
Emission implementation: os-spec-push-qed.f03 :: bremsstrahlung
block with ax_model==2 (search 'ax_model2(RW)')
```

```
Parameter container: os-axion-bremsstrahlung.f03 :: type
t_axion_bremsstrahlung_info
```

## 3.3 Compton-like axion emission (thermal bath)

The Compton-like channel samples axion production from a thermal bath via an effective local rate model. The implementation mirrors the bremsstrahlung architecture: local density proxies, Poisson macro-event sampling, optional feedback buffers, and optional evolving local temperature.

The rate is evaluated per electron using the configured coupling g_ae and scaling cs_scale, together with a proxy density field and a temperature-dependent parametrisation. As for bremsstrahlung, the emitted axion energy is sampled from a proxy distribution in w = E/T and directions are sampled isotropically.

```
Emission implementation: os-spec-push-qed.f03 :: subroutine
axion_compton_emission
```

```
Parameter container: os-axion-compton.f03 :: type
t_axion_compton_info
```

## 4. Primakoff conversion: photon ↔ axion

Primakoff conversion is implemented as a stochastic conversion operator acting on massless photon and axion species in a screened Coulomb field. The forward process converts photons to axions (γ → a) with depletion of the emitter macroparticle weight, and the inverse process converts axions to photons (a → γ) analogously.

### 4.1 Cross section and rate model

For a photon of energy ω (keV), screened by a Debye-Hückel scale κ (keV), the implemented Primakoff cross section is

$$\sigma_{\gamma\to a}(\omega) = (\alpha\, g_{a\gamma}^2\, Z^2 / 8)\, F(s)\, (\hbar c)^2, \quad s = 4\,\omega^2 / \kappa^2,$$

$$F(s) = (1 + 1/s)\, \ln(1 + s) - 1.$$

Given a target density n_t (cm^-3), the conversion rate is Γ = n_t σ c. The screening scale κ is computed from the local temperature and effective charge density via a Debye expression.

$$\Gamma(\omega) = n_t\, \sigma(\omega)\, c, \quad \kappa^2 = 4\pi\, \alpha\, (n_{eff} / T).$$

`Physics helpers: os-axion-primakoff.f03 :: primakoff_sigma_cm2, primakoff_rate_sinv, kappa_debye_keV`

### 4.2 Poisson macro-event conversion and emitter depletion

For each photon (or axion) macroparticle, the expected number of conversion macro-events over a timestep is λ = (w_emitter Γ Δt)/w_prod, where w_prod is the configured macroparticle weight of the produced species. A Poisson draw yields k conversion events. For each event, a new particle is created with weight w_prod (possibly rescaled under λ_cap), and the emitter macroparticle weight is reduced by the corresponding real-number conversion weight to conserve the total number represented by macroparticles.

`Conversion operators: os-axion-primakoff.f03 :: primakoff_photon_to_axion and primakoff_axion_to_photon`

`Poisson sampler: os-axion-primakoff.f03 :: poisson_knuth`

### 4.3 Optional photon bath initialisation

For testing and equilibrium studies, the Primakoff module optionally resamples photon momenta from a Planck distribution at a user-specified temperature. An optional quiet-start mode draws momenta in ±p pairs to reduce initial sampling noise.

`Planck resampling: os-axion-primakoff.f03 :: maybe_init_photons_planck, sample_planck_w, sample_planck_n`

## 5. Inverse processes and enforced detailed balance

Inverse (absorption) operators are implemented for thermal-bath bremsstrahlung and Compton-like channels. Rather than constructing an independent microscopic absorption cross section, the

code enforces detailed balance at the level of the local emissivity model: the absorption coefficient is derived such that, when the axion distribution equals the chosen equilibrium distribution, the net production vanishes (bin-by-bin in energy).

### 5.1 Detailed-balance construction

Let d ṅ_a/dE be the differential number emissivity (cm^-3 s^-1 (energy)^-1) implied by the forward model. Let dn_a^{eq}/dE be the equilibrium number density spectrum. Detailed balance is enforced by defining the absorption rate

$$\Gamma_{abs}(E) = [cs_{scale}\ (d\dot{n}_a/dE)] / [dn_a^{eq}/dE].$$

In a timestep Δt, each axion macroparticle is removed with probability p_abs = 1 − exp(−Γ_abs Δt), with saturation to p_abs≈1 for large optical depth.

$$p_{abs} = 1 - \exp(-\Gamma_{abs}\,\Delta t).$$

### 5.2 Bose-Einstein versus Maxwell-Boltzmann equilibrium

The equilibrium spectrum may be selected as either Bose-Einstein (BE) or the classical Maxwell-Boltzmann (MB) limit. With E in erg and w = E/(k_B T), the implemented equilibrium spectra take the form

$$dn_a^{eq}/dE = [E^2 / (2\pi^2 (\hbar c)^3)]\ [1/(\exp(w) - 1)] \quad (BE),$$

$$dn_a^{eq}/dE = [E^2 / (2\pi^2 (\hbar c)^3)]\ \exp(-w) \qquad (MB).$$

The BE option includes the expected enhancement at low energies; the MB option is appropriate when occupation numbers are uniformly small.

```
Brems absorption: os-axions.f03 :: axion_bremsstrahlung_absorb (see
'use_bose')
```

```
Compton absorption: os-axions.f03 :: axion_compton_absorb (see
'use_bose')
```

## 6. Energy and momentum exchange with the plasma

Forward emission removes energy and momentum from the plasma; absorption returns energy and momentum. The module implements a two-stage scheme: (i) during emission and absorption, per-cell deposits are accumulated in dedicated VDF buffers depE and depP; (ii) a feedback operator applies these deposits to the plasma species through a conservative affine remapping of particle momenta.

### 6.1 Deposit buffers

For each cell, depE stores the requested change in kinetic energy (in units of m_e c^2) and depP stores the requested change in momentum (in units of m_e c). By convention, emission contributes negative deposits (energy and momentum removed from plasma), whereas absorption contributes positive deposits.

`Deposit accumulation (brems emission): os-spec-push-qed.f03 :: ax_model==2 block (depE/depP updates)`

`Deposit accumulation (brems absorption): os-axions.f03 :: axion_bremsstrahlung_absorb (depE/depP updates)`

`Deposit allocation/initialisation: os-axion-bremsstrahlung.f03 :: ensure_feedback`

`Analogous Compton paths: os-axion-compton.f03 :: ensure_feedback; os-spec-push-qed.f03 :: axion_compton_emission; os-axions.f03 :: axion_compton_absorb`

### 6.2 Conservative affine feedback operator

Let a cell contain particles with momenta p_i and weights w_i (interpreted as number weights). The operator applies an affine map

$$p_i \mapsto p_i' = s\,p_i + b,$$

such that the total momentum change matches the requested deposit ΔP and, to the extent possible, the kinetic-energy change matches ΔE. The operator uses a non-relativistic kinetic-energy model E = Σ w_i |p_i|^2/(2μ), with μ = m_s/m_e for the species being updated. The map is computed cell-locally from the pre-map moments: total weight W, momentum P, and second moment U2 = Σ w_i |p_i|^2.

A uniform momentum shift per particle δ = ΔP/W produces a bulk-energy change ΔE_bulk. The remaining energy change ΔE_rem = ΔE − ΔE_bulk is applied by scaling the thermal variance about the mean by a factor s = sqrt(var_new/var), where var = U2 − |P|^2/W and var_new = var + 2 μ ΔE_rem. The corresponding affine offset is b = (1−s)⟨p⟩ + δ.

$$\delta = \Delta P/W;\ \ var = U2 - |P|^2/W;\ \ var_new = var + 2\,\mu\,\Delta E_rem;\ \ s = sqrt(var_new/var);\ \ b = (1-s)\langle p\rangle + \delta.$$

`Feedback implementation (brems): os-axions.f03 :: apply_axion_brems_feedback (internal apply_to_species)`

`Feedback implementation (Compton): os-axions.f03 :: apply_axion_compton_feedback`

## 7. Local temperature evolution and self-consistent rate evaluation

When enabled, the feedback operator estimates a cell-local temperature from the post-map thermal variance and stores it in Te_cell (keV). Subsequent emission and absorption evaluations in that cell use Te_cell instead of the constant Te_keV parameter, thereby forming a self-consistent local coupling between axion radiation and plasma thermodynamics.

The temperature estimator uses the relation for an isotropic non-relativistic distribution: ⟨|p−⟨p⟩|^2⟩ = 3 (k_B T)/(m_e c^2) in dimensionless OSIRIS units. Implementationally, Te_est = 511 keV

× (var_new/W)/3. A relaxation parameter Te_relax optionally under-relaxes the update, and bounds Te_floor_keV and Te_ceil_keV enforce numerical safety.

$T_{est,keV} = 511 \times (var_new/W)/3; \quad T \leftarrow (1-Te_relax)\, T + Te_relax\, T_{est}.$

`Temperature update: os-axions.f03 :: apply_axion_brems_feedback (Te_est/Te_new) and apply_axion_compton_feedback`

## 8. Input-deck configuration

This section documents the input-deck variables controlling the axion module. The example deck provided (test.3d) uses explicit blocks qed_axion_common, qed_axion_compton, qed_axion_bremsstrahlung, and qed_axion_primakoff. Each block maps to a Fortran namelist used by the corresponding read_input routine.

### 8.1 qed_axion_common

Shared default parameters that may be adopted by individual processes when they are not explicitly set.

| Variable | Meaning | Notes |
|---|---|---|
| Te_keV | Default plasma temperature (keV). | Applied only if a process-specific Te_keV is <= 0. |
| w_axion | Default axion macroparticle weight (number of real axions represented). | Used for variance control; process-specific override possible. |
| proba_mult | Default probability multiplier for Poisson sampling. | Used by legacy sampling paths; may be adopted by processes. |
| lambda_cap | Default cap on Poisson mean λ. | If <=0, process may apply an internal default. |

`Reader: os-particles-qed.f03 :: init_axion_common (namelist nl_qed_axion_common)`

### 8.2 qed_axion_bremsstrahlung

Configuration of axion bremsstrahlung emission and (optionally) inverse absorption.

| Variable | Meaning | Notes |
|---|---|---|
| if_bremsstrahlung | Enable axion bremsstrahlung emission. | Process is inactive if .false. |
| ax_model | Model selector. | ax_model=2 activates the thermal-plasma model |

| | | |
|---|---|---|
| | | documented here. |
| g_ae | Axion-electron coupling. | Used in emissivity; alpha_ae = g_ae^2/(4π). |
| cs_scale | Overall scale factor for rates. | Applied multiplicatively to the rate and detailed-balance absorption. |
| Te_keV | Plasma temperature (keV). | If evolve_Te=.true., Te_cell overrides this per cell. |
| include_ee | Include e-e brems contribution. | Only relevant for ax_model=2. |
| use_screening | Enable screening corrections. | A Debye-Hückel-like correction in the adopted parametrisation. |
| assume_neutral | Infer n_e from ion proxy density. | Currently the default path; ne_factor may scale it. |
| ne_factor | Multiplier applied to inferred n_e. | Allows controlled departures from strict neutrality in the proxy. |
| X_H | Hydrogen mass fraction proxy. | Enters composition factors in the emissivity model. |
| A_ion | Ion mass number. | Used to convert proxy ion density to mass density. |
| Z_ion | Ion charge state. | Used in neutrality relation and ion feedback mapping. |
| i_ion(…) | Ion species indices used for the density proxy. | Matches QED-group indexing conventions. |
| if_inverse | Enable inverse absorption via detailed balance. | Activates axion_bremsstrahlung_absorb. |
| use_bose | Choose BE versus MB equilibrium for detailed balance. | .true. selects BE; .false. selects MB. |
| w_axion | Axion macroparticle weight. | If >0, Poisson mean scales as 1/w_axion. |

| | | |
|---|---|---|
| lambda_cap | Cap on Poisson mean λ. | If exceeded, weight is rescaled to preserve expectation. |
| proba_mult | Sampling multiplier. | Legacy/auxiliary control; used when w_axion<=0. |
| apply_feedback | Accumulate depE/depP deposits. | Required for energy/momentum exchange. |
| fe_electrons | Fraction of deposits applied to electrons. | Remaining fraction is applied to ions. |
| evolve_Te | Enable Te_cell evolution from feedback operator. | Implies apply_feedback=.true. |
| Te_floor_keV | Lower bound for Te_cell. | Safety bound. |
| Te_ceil_keV | Upper bound for Te_cell. | Safety bound. |
| Te_relax | Relaxation parameter for Te_cell update. | 0: no update; 1: full update. |
| feedback_verbose | Verbose logging for feedback. | For validation/debugging. |

```
Reader/container: os-axion-bremsstrahlung.f03 ::
read_input_axion_bremsstrahlung (nl_qed_axion_bremsstrahlung)
```

```
Initialisation/linking: os-particles-qed.f03 ::
init_axion_bremsstrahlung
```

### 8.3 qed_axion_compton

Configuration of thermal-bath Compton-like axion production and (optionally) inverse absorption.

| Variable | Meaning | Notes |
|---|---|---|
| if_compton | Enable Compton-like axion emission. | Inactive if .false. |
| g_ae | Axion-electron coupling. | Used in the rate parametrisation. |
| cs_scale | Overall rate scaling. | Also applied in detailed-balance absorption. |
| Te_keV | Plasma temperature (keV). | Overridden by Te_cell if |

| | | |
|---|---|---|
| | | evolve_Te=.true. |
| assume_neutral | Infer n_e from ion proxy density. | Current default behaviour. |
| ne_factor | Multiplier applied to inferred n_e. | Proxy control. |
| Z_ion, A_ion, X_H | Composition proxies. | Enter the implemented thermal rate model. |
| i_ion(…) | Ion species indices used for proxy density. | Matches QED-group conventions. |
| if_inverse | Enable inverse absorption via detailed balance. | Activates axion_compton_absorb. |
| use_bose | BE versus MB equilibrium for detailed balance. | As in bremsstrahlung. |
| w_axion | Axion macroparticle weight. | Variance-control parameter. |
| lambda_cap | Cap on Poisson mean λ. | Saturation avoidance with weight rescaling. |
| proba_mult | Sampling multiplier. | Legacy/auxiliary control when w_axion<=0. |
| apply_feedback, fe_electrons | Deposit buffers and partitioning. | As in bremsstrahlung. |
| evolve_Te, Te_floor_keV, Te_ceil_keV, Te_relax | Local temperature evolution. | As in bremsstrahlung. |

`Reader/container: os-axion-compton.f03 :: read_input_axion_compton (nl_qed_axion_compton)`

`Emission: os-spec-push-qed.f03 :: axion_compton_emission`

`Absorption: os-axions.f03 :: axion_compton_absorb`

## 8.4 qed_axion_primakoff

Configuration of Primakoff conversion between photons and axions.

| Variable | Meaning | Notes |
|---|---|---|
| if_primakoff | Enable $\gamma \rightarrow a$ Primakoff conversion. | Acts on the photon species. |
| if_inverse | Enable $a \rightarrow \gamma$ inverse | Acts on the axion species. |

| | | |
|---|---|---|
| | conversion. | |
| g_agamma | Axion-photon coupling (GeV^-1). | Converted internally to keV^-1. |
| Te_keV | Plasma temperature (keV). | Used for κ and (optionally) photon bath initialisation. |
| use_screening | Enable screening in κ. | κ computed from Debye expression if kappa_keV<0. |
| kappa_keV | Explicit screening scale κ (keV). | <0 requests internal computation. |
| assume_neutral, ne_factor | Electron density proxy controls. | Used in κ computation. |
| include_ee | Include e-e term in screening proxy. | Affects n_eff in κ. |
| Z_ion, i_ion(…) | Target ion charge and proxy density selection. | Used to obtain n_t. |
| w_axion | Produced axion macroparticle weight. | Variance-control in conversion. |
| lambda_cap | Cap on Poisson mean λ. | Saturation avoidance. |
| debug | Enable debugging output. | For validation. |
| photon_init | Photon momentum initialisation option. | 'none' or 'planck'. |
| photon_Te_keV | Override temperature for photon bath sampling. | If >0, overrides Te_keV. |
| photon_quiet_pairs | Quiet-start sampling with ±p pairs. | Reduces noise in initial photon bath. |
| photon_nmax | Series truncation parameter for Planck sampling. | Controls sampling accuracy and cost. |

```
Info container: os-axion-primakoff-info.f03 :: type
t_axion_primakoff_info (read_input)
```

```
Conversion kernels: os-axion-primakoff.f03 ::
primakoff_photon_to_axion / primakoff_axion_to_photon
```